

# Naveen Prabakar

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## EDUCATION

### Iowa State University

Bachelor of Science in Computer Science, Minor in Data Science & Applied AI

Ames, IA

Expected May 2026

- **GPA: 3.85/4.00**, Dean's List (01/23 – Present)
- **Certifications:** Prompt Design in Vertex AI (Google Cloud), AI Fundamentals (DataCamp)

## WORK EXPERIENCE

### AI Software Engineering Intern

09/25 – Present

Merrill Manufacturing

Remote

- Engineered a full-stack AI assistant using React, Flask (Python), SuiteScript, PostgreSQL, and NetSuite's **LLM module**, enabling employees to query live ERP data in natural language.
- Built a **multi-stage Agentic AI pipeline** with Retrieval-Augmented Generation (**MiniLM + FAISS**) and a PostgreSQL conversational memory layer for context-aware interactions.
- Implemented **DeBERTa-v3-based BERT guardrails** to detect and block prompt-injection and jailbreak attempts, enhancing reliability and compliance.
- **Reduced response latency by 60%** through parallelized reasoning batches, semaphore-based concurrency control, and optimized LLM agent orchestration.

### Database Administrator Intern

05/25 – 08/25

Uline

Pleasant Prairie, WI

- Optimized **2–3 SQL Server queries weekly**, achieving up to **42% reduction** in execution times, collaborating with the Senior DBA on performance tuning.
- Created **stored procedures** to detect log reader agent failures and partnered with the ESM team to integrate alerts into Nagios, improving replication visibility.
- Handled **20+ database operations** including permission management, backups and UAT restores, stored procedure deployments across SYS/UAT/PROD, SSRS uploads, and automated tasks using **VBScript**.

## PROJECTS

### [Crypto Analytics Platform](#) | Go, React, TypeScript, Cassandra, Docker, AWS EC2

06/25 – 08/25

- Built scalable microservices backend using Go and Cassandra, processing **10k+ records/week** and enabling sub-second queries on time-series crypto data.
- Developed **automated ETL pipelines** and real-time dashboards on AWS, ensuring **99.9% uptime** and supporting multi-cloud scalability.

### [School Subject Assistant](#) | Python, Redis, Pinecone, JavaScript, HTML, CSS, AWS, Docker

08/25 – 09/25

- Built full-stack AI Q&A system for statistics using 1,000 PDF pages, **Pinecone vector search**, and **OpenAI embeddings** for context-aware solutions.
- Deployed **serverless architecture** (Docker → AWS ECR → AWS Lambda → API Gateway) with Redis **session memory**, achieving minimal downtime; approximately 10 seconds per cold start

### [E-commerce Phone Store](#) | React, JavaScript, Node.js, Express.js, MongoDB

03/25 – 05/25

- Built full-stack e-commerce web app, implementing core features including **product listing, cart management, and checkout workflows**.
- Developed responsive interfaces with **form validation** and engineered backend **REST APIs** with MongoDB integration to handle authentication, product posting, and real-time cart updates.

### [AI-Powered Note-Taking App](#) | Java, Spring Boot, MySQL, WebSockets, CI/CD

08/24 – 12/24

- Built backend infrastructure using Spring Boot and MySQL, **optimizing data retrieval** by 20%.
- Integrated OCR functionality with Tesseract API, achieving 90% text extraction accuracy.
- Developed **real-time** live editing features via WebSockets, allowing multi-user sessions.

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL, JavaScript, TypeScript, Go, HTML, CSS

**Databases:** MySQL, MongoDB, SQL Server, Cassandra, Redis, Pinecone, PostgreSQL

**Cloud/DevOps:** Docker, CI/CD, AWS (S3, EC2, ECR, Lambda, API Gateway)

**Frameworks/Tools:** SpringBoot, React, Express, Node, Flask, SuiteScript